Project - Low Level Design

on

Construction Planning Assistant Agent

Course Name: Agentic AI

Medicaps University – Datagami Skill Based Course

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Introduction

1.1 Scope of the Document

This document describes the Low-Level Design of the Construction Planning Assistant Agent built using Python, LangChain/CrewAI, and LLM APIs. The application acts as an intelligent agent that generates construction plans by breaking down high-level goals into structured tasks and schedules.

This document covers:
• Internal system components (Planner Agent, Task Decomposer, Tool Interface, Scheduler)
• Agent workflow and reasoning process
• API structure and data flow
• Integration with LLM and external tools (mock resource checker)
• Task dependency handling and execution planning

1.2 Intended Audience

This document is intended for:
• Developers working on the application using Python, LangChain/CrewAI, and LLM APIs to understand system implementation
• Reviewers to evaluate the design and working of the agent
• Team members involved in development and maintenance
• Personnel handling integration with tools and APIs (like resource checker and LLM)
• Contributors managing deployment and updates of the system

This document provides a clear understanding of the system's internal design, agent workflow, and how construction plans are generated.

1.3 System Overview

The Construction Planning Assistant Agent is an intelligent AI-based system designed to help in construction planning. It takes a high-level input from the user, such as project goals or requirements, and converts it into a clear and structured plan.

The system works step by step. First, it understands the user input using an LLM (Large Language Model). Then, it breaks the main goal into smaller and manageable tasks. After that, it checks the availability of required resources using tool interfaces (mock tools).

The agent also understands task dependencies, meaning it decides which task should be done first and which can be done later. Based on this, it organizes all tasks in the correct order.

Finally, the system generates a detailed execution schedule that helps in completing the construction work efficiently.

The application is built using Python, LangChain/CrewAI, and LLM APIs, which allow the agent to perform reasoning, decision-making, and task planning automatically.

2. System Design

2.1 Application Design

The Construction Planning Assistant Agent is designed using a modular architecture, where each component has a specific role. This makes the system easy to understand, maintain, and extend.

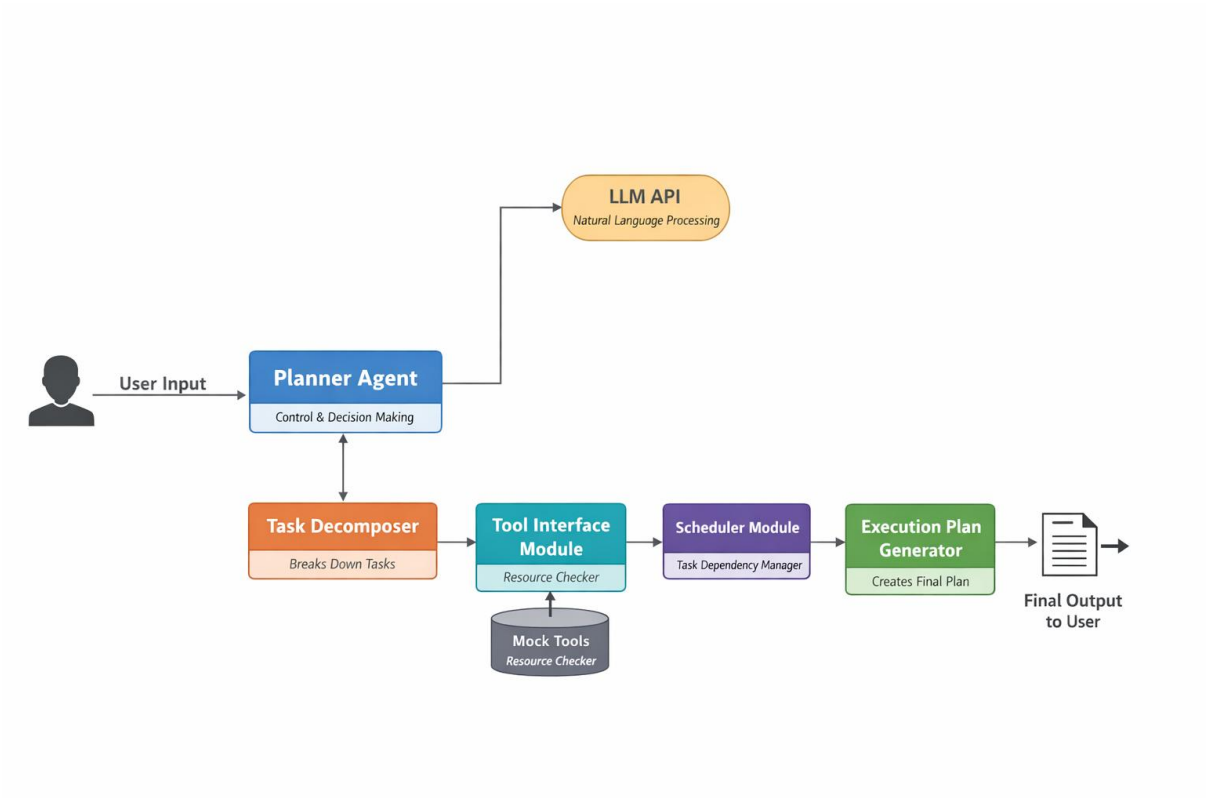
The main component is the Planner Agent, which controls the overall workflow. It receives user input and coordinates all other components.

The Task Decomposer breaks the high-level goal into smaller tasks. The Tool Interface Module is used to check resource availability and interact with external or mock tools. The Scheduler Module organizes tasks based on dependencies and creates a proper execution order.

All components work together in a step-by-step process to generate an accurate construction plan. The system uses LangChain/CrewAI to manage agent behavior and LLM APIs for understanding and reasoning.

This design ensures flexibility, scalability, and efficient task planning.

System Architecture:



Architecture Explanation:

- User gives input (construction goal)
- Planner Agent controls the whole process
- LLM API helps to understand the input
- Task Decomposer breaks goal into small tasks
- Tool Interface checks resources using mock tools
- Scheduler arranges tasks in correct order
- Execution Plan Generator creates final plan
- Final output is sent to the user.

2.2 Process Flow

1. User gives high-level input (project goal or requirement)
2. Planner Agent receives and understands the input using LLM
3. Task Decomposer breaks the goal into smaller tasks
4. Tool Interface checks resources and required data
5. Scheduler arranges tasks based on dependencies
6. System generates final construction plan/schedule
7. Output is shown to the user

2.2 Process Summary

The system takes a high-level input from the user and processes it using the Planner Agent. It breaks the input into smaller tasks, checks resources, and arranges tasks in the correct order. Finally, it generates a complete construction plan or schedule as the output.

2.3 Information Flow

The information in the system flows step by step between different components.

First, the user provides input to the Planner Agent. The agent sends this input to the LLM for understanding. Then, the processed data is passed to the Task Decomposer, which breaks it into smaller tasks.

After that, task details are sent to the Tool Interface Module to check resources and required information. The validated data is then passed to the Scheduler Module, which arranges tasks in the correct order.

Finally, the complete plan is generated and sent back to the user as the output.

2.4 Internal Flow

• Planner Agent receives user input
• LLM processes and understands input
• Task Decomposer creates smaller tasks
• Tool Interface checks resources
• Scheduler arranges tasks in order
• Final plan is generated.

2.5 Output Flow

• System prepares final construction plan output
• Tasks are arranged in proper logical order
• Schedule with dependencies is properly created
• Output is formatted in clear readable format
• Final result is sent back to user.

2.4 Components Design

The system is divided into different components, where each component has a specific role.

• Planner Agent – Controls overall workflow and manages decision-making process
• Task Decomposer – Breaks high-level goals into smaller manageable tasks
• Tool Interface Module – Checks resource availability and interacts with external tools
• Scheduler Module – Arranges tasks in proper order based on dependencies
• LLM Integration Module – Understands input and supports reasoning process

All these components work together to generate an efficient construction plan.

2.6 API Catalogue

The system provides APIs to interact with the Construction Planning Assistant Agent and generate construction plans.

• /generate-plan – Accepts user input and generates a construction plan
• /decompose-task – Breaks high-level goals into smaller tasks
• /check-resources – Checks availability of required resources
• /schedule-tasks – Arranges tasks based on dependencies
• /get-output – Returns the final execution plan to the user

These APIs help in smooth communication between the user and the system.

1. References

This document is prepared with the help of the following resources:

• LangChain/CrewAI official documentation
• LLM API documentation (OpenAI or similar)
• Python official documentation
• Online tutorials and learning resources for Agentic AI
• Project-related research and design materials

These references helped in understanding and designing the system.